
Churn Decision Intelligence System

Overview

This project operationalizes a customer churn prediction system into a decision-oriented ML product.

The analytical phase of the project focused on identifying economically efficient retention strategies through churn modeling, threshold analysis, intervention simulation, and expected value evaluation. Rather than treating churn prediction as a standalone classification task, the project framed the model as a business decision engine designed to support targeted retention actions under real-world cost constraints.

The product layer extends this framework into a deployable system by exposing the decision pipeline through a FastAPI service and an automated n8n workflow.

The system supports:

- Real-time and batch customer scoring
- Automated risk segmentation
- Action recommendation generation
- Expected value estimation
- Workflow-based operational automation

Instead of returning prediction probabilities alone, the system produces decision-ready outputs that can directly support CRM, retention, and marketing operations.

This project demonstrates the transition from predictive modeling to operational machine learning systems aligned with business decision-making.

1. Overview

This project extends a traditional customer churn prediction model into a prototype decision-support system for customer retention. Rather than stopping at churn probability estimation, the system translates model predictions into economically-aware retention actions designed to support operational business decisions.

The project combines machine learning, API development, and workflow automation into a unified pipeline capable of processing both individual customers and batch customer datasets.

At a high level, the system performs the following steps:

Customer Data

→ Churn Prediction

→ Risk Segmentation

→ Retention Policy Selection

→ Expected Value Evaluation

→ Actionable Business Output

The system is designed around a core principle:

Predictive models create value only when predictions are transformed into decisions.

To support this objective, the project introduces a policy layer that maps churn probabilities to retention strategies based on estimated business impact and intervention cost.

Examples of generated actions include:

- Premium retention offers for high-risk customers
- Targeted email campaigns for medium-risk customers
- Monitoring-only decisions for low-risk customers

The project also includes:

- A REST API built with FastAPI for real-time and batch inference
- Internal preprocessing and feature engineering pipelines
- Automated workflow orchestration using n8n
- Tracking and logging fields for traceability and operational monitoring

The overall goal is to demonstrate not only how to train a predictive model, but also how to operationalize machine learning outputs within a realistic business decision framework.

2. System Architecture

The project is designed as an end-to-end customer retention decision pipeline that combines machine learning inference, business decision logic, and workflow automation. At a high level, the system processes customer data through several sequential layers:

Input Data

- Preprocessing & Feature Engineering
- Churn Prediction Model
- Risk Segmentation
- Retention Policy Layer
- Expected Value Evaluation
- Action Output
- Automation Workflow

Architecture Components:

Layer	Responsibility
Input Layer	Accept raw customer data in JSON or CSV format
Preprocessing Layer	Apply encoding and feature engineering internally
ML Inference Layer	Generate churn probability predictions
Segmentation Layer	Convert probabilities into business risk categories
Policy Layer	Map risk levels to retention strategies
Cost Evaluation Layer	Estimate expected business value of interventions
Automation Layer	Route decisions through automated workflows using n8n

The system intentionally separates prediction from decision-making. Instead of treating churn probability as the final output, the pipeline translates model predictions into operational retention actions informed by business rules and economic considerations.

To support traceability and monitoring, prediction outputs also include operational metadata such as timestamps, request IDs, and model version identifiers.

3. Business Decision Layer

The project extends beyond churn prediction by converting model outputs into economically-aware retention decisions.

Customers are segmented based on predicted churn probability:

Probability Range Segment

≥ 0.75	High Risk
0.50 - 0.75	Medium Risk
0.30 - 0.50	Low-Medium Risk
< 0.30	Low Risk

Each segment is mapped to a predefined retention strategy:

Segment	Example Action
High Risk	Premium retention offer / personal outreach
Medium Risk	Targeted email campaign
Low-Medium Risk	Experimental targeting
Low Risk	No action

To avoid economically inefficient interventions, the system estimates expected value before assigning actions:

$$\text{Expected Value} = (\text{probability} \times \text{retention_success_rate} \times \text{customer_value}) - \text{intervention_cost}$$

If the projected value of an intervention is negative, the system overrides the recommendation to: *no action*

This decision layer separates prediction from business strategy, enabling the system to support operational retention workflows rather than prediction alone.

4. API Design

The system is exposed through a REST API built with FastAPI, enabling both real-time and batch customer inference workflows.

The API is designed to accept raw customer data while handling preprocessing, feature engineering, and decision logic internally.

Endpoint	Description
GET /	Health check endpoint
POST /predict	Predict churn risk for one or multiple customers using JSON input
POST /predict-file	Perform batch prediction using CSV input

The screenshot displays the Swagger UI for the API. It shows two endpoints: a GET endpoint for a health check and a POST endpoint for batch prediction using CSV input.

Endpoint 1: GET /

- Parameters:** No parameters.
- Request body:** Not applicable for GET.
- Responses:**
 - 200 Successful Response:** Media type: application/json. Example Value: A JSON object with fields like credit_score, age, tenure, etc.
 - 422 Validation Error:** Media type: application/json. Example Value: A JSON object with an 'detail' field containing an array of error messages.

Endpoint 2: POST /predict-file

- Parameters:** No parameters.
- Request body:** multipart/form-data. Field: file (required, string).
- Responses:**
 - 200 Successful Response:** Media type: application/json. Example Value: A JSON object with a 'detail' field containing an array of error messages.
 - 422 Validation Error:** Media type: application/json. Example Value: A JSON object with a 'detail' field containing an array of error messages.

Output Structure

Each prediction returns decision-oriented outputs rather than probability alone.

5. Automation Workflow (n8n)

The project integrates n8n to automate operational decision workflows after model inference.

The automation layer connects customer prediction outputs with downstream business actions, enabling the system to function as an end-to-end retention pipeline rather than a standalone ML model.

Workflow Overview

CSV / JSON Input

→ API Request

→ Prediction & Decision Layer

→ Action Routing

→ Result Logging / Export

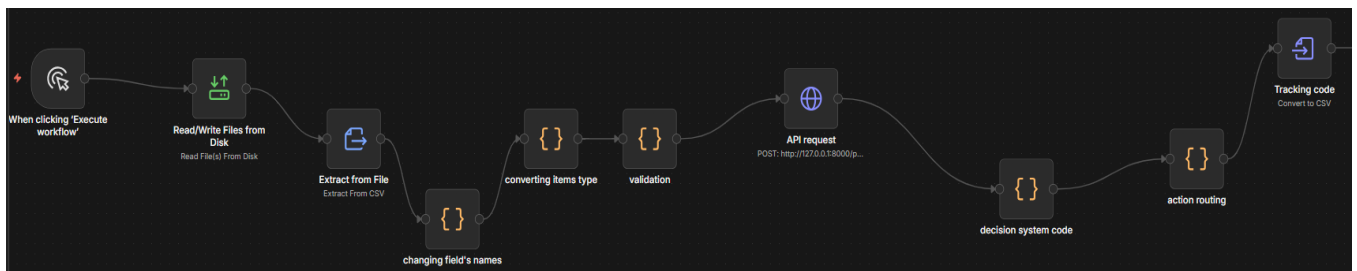
Automated Actions

Based on the predicted retention strategy, the workflow can route customers into different operational paths:

Action	Example Workflow
Premium Retention Offer	Send customer to CRM / sales team
Email Campaign	Trigger automated marketing flow
No Action	Log prediction results only

The workflow also supports batch CSV processing, allowing large customer datasets to be evaluated automatically through the API pipeline.

This automation layer demonstrates how predictive systems can be integrated into operational business processes using low-code orchestration tools.



6. Input & Output Structure

The API accepts raw customer-level data in either JSON or CSV format.

Expected input features include:

- CreditScore
- Geography
- Gender
- Age
- Tenure
- Balance
- NumOfProducts
- HasCrCard
- IsActiveMember
- EstimatedSalary
- Complain
- Satisfaction Score
- Card Type
- Point Earned

All preprocessing steps — including encoding and feature engineering — are handled internally within the inference pipeline to ensure consistency between training and production environments.

The system returns decision-oriented outputs rather than prediction probabilities alone.

Example response:

```
{
  "probability": 0.82,
  "segment": "high_risk",
  "action": "premium_retention_offer",
  "expected_value": 120.5
}
```

Depending on the endpoint, outputs are returned as:

- JSON responses for real-time automation workflows
- CSV files for batch analysis and reporting

Basic validation rules are applied before inference, including range checks, binary validation, and categorical consistency validation.

7. Project Structure

Folder	Purpose
data/	Source datasets used during development
models/	Serialized ML models and preprocessing artifacts
src/	Core API, inference, and decision pipeline
notebooks/	Exploratory analysis and experimentation
test_data/	Example input files for testing
workflow/	n8n workflow exports
outputs/	Generated prediction and decision outputs

```
product/
├── data/
│   └── churn.csv
├── models/
│   ├── model.pkl
│   ├── encoder.pkl
│   └── feature_engineer.pkl
├── src/
│   ├── app.py
│   ├── config.py
│   ├── predict.py
│   ├── encoding.py
│   ├── feature_engineering.py
│   └── decision_logic.py
├── notebooks/
│   └── churn_analysis.ipynb
├── workflow/
│   └── Churn prediction simulator.json
├── outputs/
│   └── prediction_results.csv
└── README.md
```

8. Future Improvements

Several extensions could further improve the system and move it closer to a production-grade retention platform:

- Integration with CRM and marketing platforms for fully automated intervention workflows
- Real-time monitoring and logging dashboards for prediction tracking
- Model retraining and drift detection pipelines
- Authentication and rate limiting for secured API access
- Containerization and cloud deployment
- A/B testing framework for evaluating retention strategies
- Dynamic policy optimization based on observed intervention outcomes
- Customer lifetime value estimation for more accurate expected value calculations

The current implementation focuses on demonstrating how machine learning predictions can be operationalized into actionable business decisions through APIs and workflow automation.